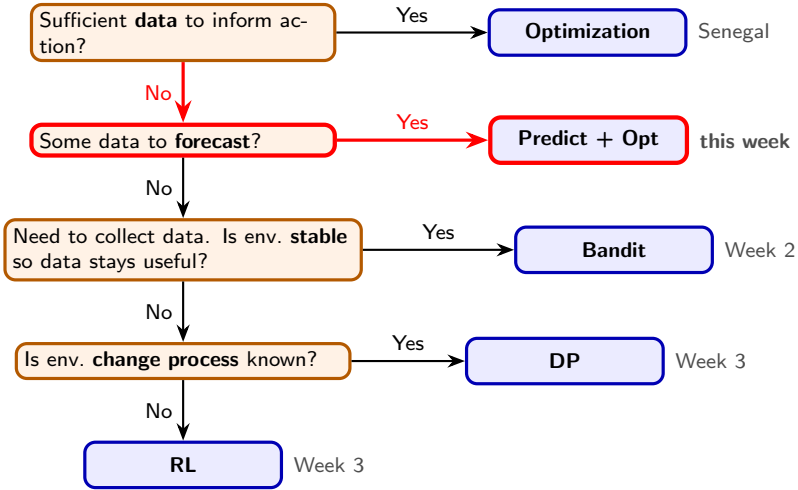
Lecture 7: *Smart* "Predict, then Optimize"

Integrating prediction and optimization.

Peter Zhang

12768/19867/94867: From Data to Action

A Very Likely Data-to-Action Path in Policy Analytics



Problem Definition

The state opioid response office can fund a few overflow clinics.

- Six counties, each already treating **7** a year. Anyone past that is turned away.
- A clinic costs **1** to run, whether we use it or not. Turning people away costs **4**.
- Commit in **September**; demand is known only in **December**.

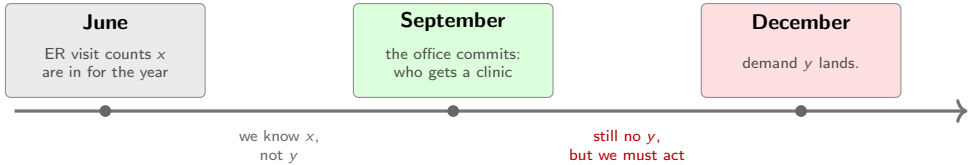
Last year's six counties

county	x: ER visits *last* year, to June	y: demand *last* December
1	1	2
2	2	5
3	3	8
4	4	10
5	5	11
6	6	12

1. If you knew y ahead of time (e.g., in September), which counties would you open a clinic in?
2. If you can only observe x by the time you make decisions, which counties would you open a clinic in?
3. How do you measure the quality of your decision? In other words, how would you compare two different decisions?
4. What has to be true for the best decision generated from this year to work next year?

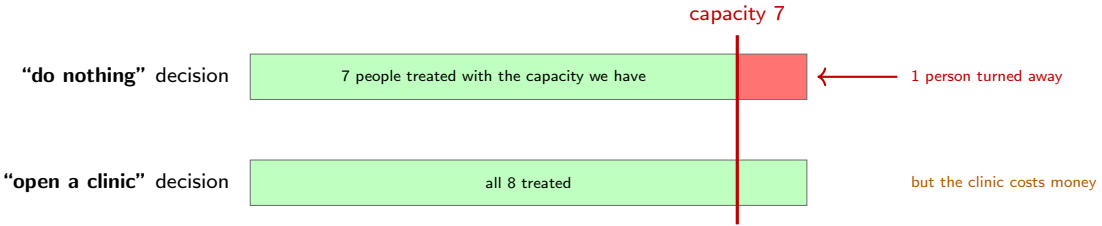
Sequence of Events

The **state opioid response office** holds a small fund for overflow treatment clinics. Six counties. Each already treats **7** people a year.



What is the Decision for Each County?

Forget the prediction part for a moment. Take one county, and suppose we know exactly **8** people will come looking for treatment via some magical oracle device. For example, two decisions have different costs.



Total Cost

For each county i, choose $\text{open}_i \in \{0, 1\}$ to minimize

$$\sum_i \text{cost}_i, \qquad \text{cost}_i = \begin{cases} 1 & \text{if open}_i = 1 \quad \text{(the clinic costs this, needed or not)} \\ 4 & \text{if open}_i = 0 \text{ and } y_i > 7 \quad \text{(we turned people away)} \\ 0 & \text{if open}_i = 0 \text{ and } y_i \leq 7 \end{cases}$$

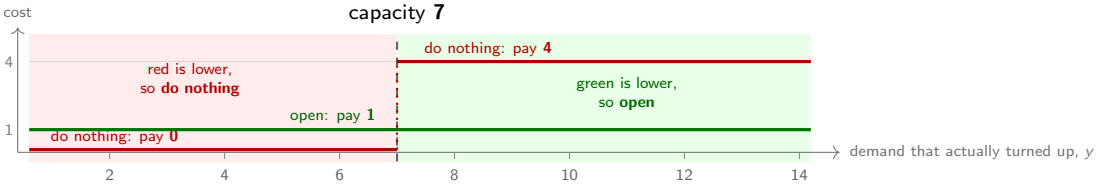
x_i is known in September. y_i is not.

Simplifying assumptions (in order to isolate our discussion to be on prediction-decision)

- The six decisions **do not have interactions**, i.e., each county has its own independent budget, and there's no state budget to be fought over.
- The prices are **fixed numbers** and given to us already. The "turning away" cost is not dependent on the number of people we turn away.

Again ... If We Had an Oracle

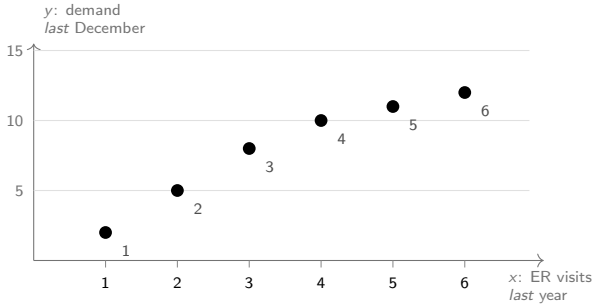
Hypothetically, suppose we already knew how many people would show up.



Then it is easy. Below 7 doing nothing is cheaper, above 7 opening the clinic is cheaper.

Now Without Oracle, We Rely on Past Data

This is **last year**'s data from all six counties. We could see a county's ER visits in **June**! Before the September decision point, and before the December realization of the demand.



county	x	y
1	1	2
2	2	5
3	3	8
4	4	10
5	5	11
6	6	12

Aside: Mixing these Six Data Points Together as One Dataset Requires Assumptions

Fitting *one* prediction model to all six data points basically says that county 6's experience tells us something useful about county 1. That is not automatically true, unless...

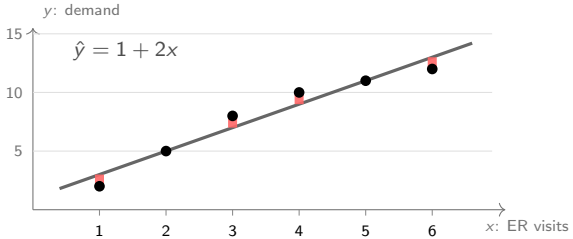
1. **Same mechanism.** A given ER-visit rate means the same thing in every county. They differ in size and in x, and are otherwise alike.
2. **Comparable units.** Both columns are in the same units, e.g., "per 10,000 residents".
3. x **carries what matters.** Anything else moving demand is common to all six, or unrelated to x.
4. **The misses are noise.** We treat $-1, 0, +1, +1, 0, -1$ as unpredictable rather than as a pattern we skipped.
5. ...

But these details aside...

Fit a Prediction Model on Last Year's Data

Grant those (plus one more assumption: that next year behaves like last year). What we want now is a **prediction model**, a rule that turns June's x into a guess at December's y and *explains* the relationship of the dots.

So what counts as a good explanation? Typical prediction/training/stats/ML literature work with something like **mean squared error**: square each discrepancy, add them up.

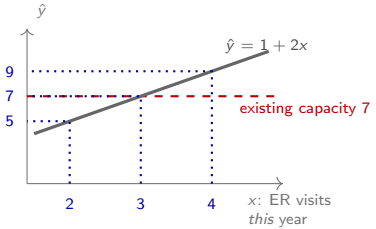


county	$\hat{y}$	miss
1	3	-1
2	5	0
3	7	$+1$
4	9	$+1$
5	11	0
6	13	-1

By that metric the winner is exactly $\hat{y} = 1 + 2x$, with squared discrepancies between the dots and the fitted line adding to **4**.

This Year: We Now Have the Line; and Early Signals x

This September the six counties report their ER visits, and December has not happened.



ER visits this year, x	2	3	4
Counties reporting it	1 and 2	3 and 4	5 and 6
what the line predicts, $\hat{y}$	**5**	**7**	**9**
Decision	no clinic	no clinic	**open**

Two Kinds of Loss

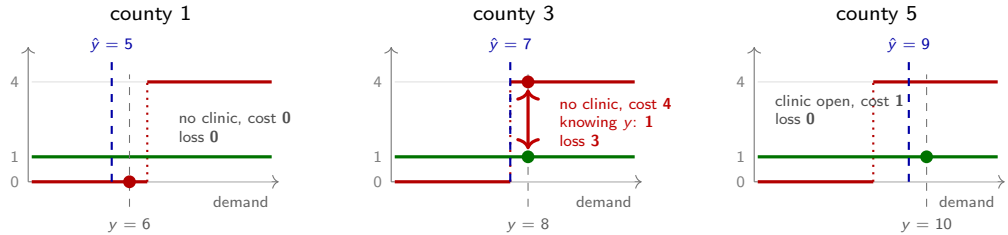
	the discrepancy it measures	**here**
prediction loss	the predicted value against the true value	$(\hat{y} - y)^2$
decision loss	the cost of the decision we made against the cost of the decision we *would* have made knowing y	$c\big(z^\star(\hat{y}),\, y\big) \;-\; \min_z c(z, y)$

December Arrives

county	x	$\hat{y}$	**actual** y	our cost	knowing y	**decision loss**
1	2	5	6	0	0	0
2	2	5	4	0	0	0
3	3	7	**8**	**4**	1	**3**
4	3	7	6	0	0	0
5	4	9	10	1	1	0
6	4	9	8	1	1	0
			total	**6**	3	**3**

The Same Three, Drawn

Blue: prediction. Grey: true demand. Red gap: decision loss.



Separating Two Issues

Two different things have been going wrong.

1. **We do not know the future.** (Technical term: the *data generating process* might change) December surprises us. Even a flawless model would be surprised, because the information simply is not there in September.
2. **We fit the line to sit close to data points, not to produce good decisions.** That is a choice we made about the loss function, and it has nothing to do with uncertainty.

We want to eliminate the second problem, knowing that the first problem is irreducible and may interact with the second problem.

Assuming Next Year's Randomness = Last Year's Randomness

We can score a line by prediction loss. What if we score it by decision loss as well?

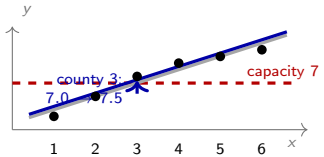
1. Run the line on every county for a prediction $\hat{y}$.
2. Apply the rule: open a clinic wherever $\hat{y} > 7$.
3. Charge each county: **1** if we opened, **4** if we did not and demand went over, **0** otherwise.
4. Subtract what the same county would have cost had we known y.

county	$\hat{y} = 1 + 2x$	open?	y	needed?	our cost	knowing y	loss
1	3	no	2	no	0	0	0
2	5	no	5	no	0	0	0
3	7	no	8	**yes**	**4**	1	**3**
4	9	yes	10	yes	1	1	0
5	11	yes	11	yes	1	1	0
6	13	yes	12	yes	1	1	0
				total	**7**	4	**3**

The best-fitting line from previous prediction step has a **decision loss of 3**.

Is it Possible to Improve Decision Loss from the Best-Fitting Line?

Let's say we try another line $\hat{y} = 1.5 + 2x$. Every prediction gets worse.



county	$\hat{y} = 1.5 + 2x$	open?	y	needed?	our cost	knowing y	loss
1	3.5	no	2	no	0	0	0
2	5.5	no	5	no	0	0	0
3	**7.5**	**yes**	8	yes	**1**	1	**0**
4	9.5	yes	10	yes	1	1	0
5	11.5	yes	11	yes	1	1	0
6	13.5	yes	12	yes	1	1	0
				total	**4**	4	**0**

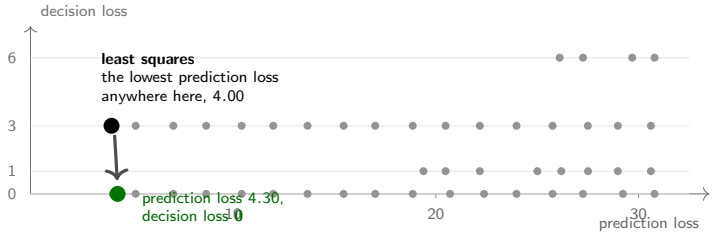
Decision loss 0.

What We Just Traded

	prediction loss	**decision loss**
$\hat{y} = 1 + 2x$, the best fit there is	**4.0**	3
$\hat{y} = 1.5 + 2x$	5.5	**0**

We paid **1.5** in prediction loss and got all **3** back in decision loss.

Try Other Lines



Summarizing The Lecture in Two Lines

Three objects, all of which we have been using already.

$\hat{y} = f_\theta(x)$	the prediction model	a line, a tree, neural net
$z^\star(\hat{y}) = \arg\min_z c(z, \hat{y})$	the decision	what the optimizer returns
$c(z, y)$	the cost	what decision z costs when the truth is y

Ordinary training:

$$\min_\theta \sum_i \ell\big(f_\theta(x_i),\ y_i\big)$$

Decision-focused training:

$$\min_\theta \sum_i c\big(z^\star(f_\theta(x_i)),\ y_i\big)$$

Mapping the Previous Example in These Symbols

x	the county's ER overdose visits, known by June
y; $\hat{y}$	the treatment demand that realizes in December; predicted demand
f_θ	the line $\theta_0 + \theta_1 x$, so θ is the intercept and slope
z	open a clinic, or do not
$c(z, y)$	1 if z is open; 4 if z is do-nothing and $y > 7$; 0 otherwise
decision loss	$c(z^\star(\hat{y}), y) - \min_z c(z, y)$
$z^\star(\hat{y})$	open exactly when $\hat{y} > 7$, because that is the cheaper option *if you believe* $\hat{y}$
$\ell(\hat{y}, y)$	$(\hat{y} - y)^2$

Shifting the line from $1 + 2x$ to $1.5 + 2x$ moved θ_0 by 0.5. Prediction loss went up; decision loss went to zero.

Exercise

The state has **5,000 naloxone kits** to send out before winter, across **40 counties**.

- Sending s kits to county i reverses about $v_i\, s - \gamma_i s^2$ overdoses, with γ_i known from past distributions.
- v_i, what the *first* kit buys, is **not** known. It is predicted from four features: last year's overdose deaths, EMS response time, treatment slots per capita, and pharmacy density.
- The prediction model is a **gradient-boosted tree**, not a line.
- The kits must all go somewhere, and no county can receive a negative number.

Spend a few minutes: Write down what each symbol is here.

$x_i =$ _____ $\quad y_i =$ _____ $\quad f_\theta =$ _____ $\quad z =$ _____

$c(z, y) =$ _____

What does $z^\star(\hat{y})$ *represent*?

Putting Some Uncertainty Back

Everything since we split the two problems has assumed next year repeats last year. Drop that now.

With the future assumed known, we minimized

$$\min_{\theta} \sum_{i} \Big[c\big(z^{\star}(f_{\theta}(x_i)),\, y_i\big) \;-\; \min_{z} c(z, y_i) \Big]$$

In general (x, y) is drawn from a distribution we do not control, so what we want is

$$\min_{\theta} \; \mathbb{E}_{(x,y)} \Big[c\big(z^{\star}(f_{\theta}(x)),\, y\big) \;-\; \min_{z} c(z, y) \Big]$$

The sum over past counties *estimates* that expectation, and is only as good as the assumption that past counties resemble future ones.

$z^{\star}$ should also be revised, as $\arg\min_z \mathbb{E}\big[c(z, y) \,\big|\, x\big]$.

Additional Questions

Can't we just re-weight the prediction training loss function to reflect decision loss?